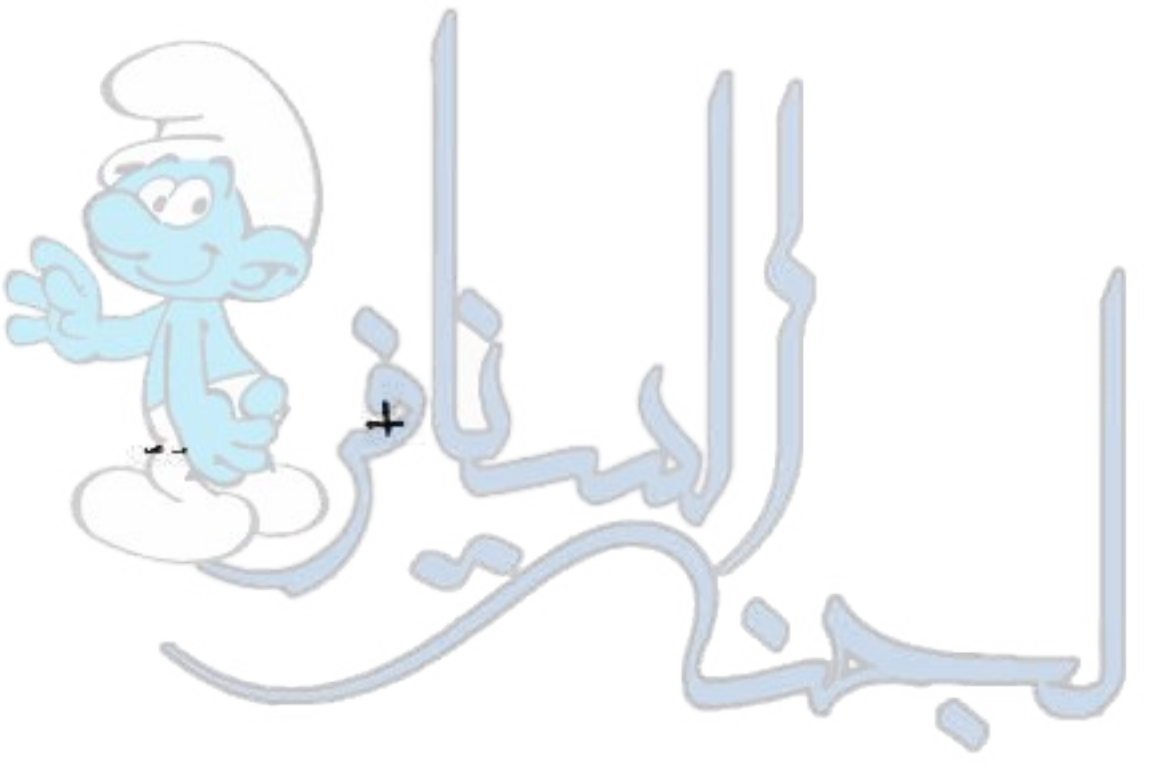
Q1: Use bisection to determine the root of $f(x) = \cos(e^x - 2)$ within $\varepsilon_s =$ 3%. Employ initial guesses of $x_L = 0.5$ and $x_u = 1.5$



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Q2: Based on Newton-Raphson method, solve the problem up to five iterations to find the root of $f(x) = x^3 - 2x^2 - 5$. Use $x_0 = 5$ as initial value.



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Q3: Employ LU decomposition method to determine the first column of the matrix invers $[A]^{-1}$ for the following system of linear equations;

$$10x_1 + 2x_2 - x_3 = 27$$

$$-3x_1 - 6x_2 + 2x_3 = -61.5$$

$$x_1 + x_2 + 5x_3 = -21.5$$



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Q4: Use Gauss-Seidel method to solve the following system until the present relative error falls below $\varepsilon_s = 5\%$. Use $x_1 = x_2 = x_3 = 0$ as initial values.

$$2x_1 - 6x_2 - x_3 = -38$$

$$-3x_1 - x_2 + 7x_3 = -34$$

$$-8x_1 + x_2 - 2x_3 = -20$$

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